

Executive Summary: Temporal Network Signals and Cross-Disciplinary Concept Spread

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1 Research Question and Scope

This five-iteration run investigated whether temporal changes in how scientific concepts connect to other topics predict whether those concepts will spread across disciplinary boundaries. The study used the OpenAlex bulk snapshot (476 million works) and tracked 12,499 concepts across 27,393 adoption episodes in 26 fields. Two research questions were addressed:

RQ1 Which early network indicators predict cross-field breadth?

RQ2 How do concepts diffuse across fields—which fields come next?

2 Key Findings

2.1 Finding 1: Open early neighbourhoods predict broader reach

Concepts whose early co-occurrence networks acquire partners from diverse communities (high novelty, low persistence) later reach more fields. The OPEN index—a composite of six ego-network indicators—was confirmed, marginally, on a fresh 2015–2017 cohort (intervals include zero at the strictest control rungs) and partially confirmed on a vocabulary-free Frame N population. The descriptive pool over six non-selection bodies is:

$$\text{OPEN}_{\text{home}} = +0.069 \text{ } [+0.038, +0.100], \quad I^2 = 0, \quad 6/6 \text{ positive.}$$

The measurable signal behaves like static topical non-redundancy: a within-concept permutation null absorbs the year-to-year component (+0.008, null), while fixed- n rarefaction retains 68% and degree-preserving nulls retain 91–100%. The temporal test is too unreliable (split-half 0.01–0.05; planted churn not recovered) to rule out partner turnover. OPEN adds no practical prediction over a five-feature popularity baseline (+0.002 [−0.003, +0.008]).

2.2 Finding 2: Seven of ten held-out indicators confirmed

From 53 early network indicators screened on development fields, the top 10 were tested on four held-out field groups (DerSimonian–Laird pooling, Holm correction). Seven confirmed for rarefied breadth (O_{2r} , $m = 50$):

Caveat: the two strongest (M0_density_end, D_vol_end) use cumulative field history from 1995; post-onset rescaling halves their effects (attenuation ~ 0.50).

Indicator	Family	Pooled PSP	Confirmed?
M0_density_end	Frontier	+0.375	Yes
D_vol_end	Frontier	+0.307	Yes
CONTACT_REACH	Frontier	+0.211	Yes
n_{comm}	Ego-net	+0.167	Yes
NOV _{res}	Ego-net	+0.151	Yes
RETENTION_RATIO	Frontier	−0.114	Yes
ego_density	Ego-net	−0.102	Yes

2.3 Finding 3: The consistency–breadth reversal

Cheng et al.’s (2023) ideational consistency predicts next-year volume (+53.5%/SD), but once current size is controlled, only +1.3% remains. As an early trait, consistency predicts *narrower* later breadth:

$$\text{PSP} = -0.069 \text{ } [-0.093, -0.047], \quad 5/5 \text{ groups negative, } I^2 = 0.$$

Replicated on the fresh cohort (−0.111); not confirmed on Frame N (−0.064, CI includes 0). Consistency $\approx$ weighted edge persistence (Spearman 0.77).

2.4 Finding 4: Concepts spread to fields related to their retained fields

Concepts enter next the fields related to those currently retaining them. On the held-out pool of an independent frame (3,162 concepts, 6,978 entry events):

$$d_0 = 0.322 \text{ } [0.291, 0.355], \quad \text{LR} = 325.8 \text{ } (p < 10^{-70}).$$

This extends the principle of relatedness from economic complexity. However, the volume-matched contrast is null (−0.028), and under min-conditional-probability proximity the effect reverses ($d_0 = -0.021$). Verdict: PARTIAL.

2.5 Finding 5: Breadth gap is 73% early contact, 25% retention

An exact decomposition ($\log B_n = \log E_2 + \log M + \log \rho$) of the top-vs.-bottom breadth tercile gap shows that, on the held-out pool (Medicine excluded, $n = 1,825$), 73% is early contact diversity (E_2) and 25% is retention (ρ); frontier advance (M) is near zero. Broad concepts start with wider contact, not by advancing a wider frontier. The shares are an accounting identity, not causal effects.

3 What Did Not Work

Five concept-level candidates were screened and closed:

- **Naturalisation gap** A_h^* : $\Delta\rho = -0.006$, 0/4 groups positive.
- **Co-occurrence diversity** D_{ratio} : $\Delta\rho = +0.006$, CI spans zero.
- **Frequency-free selectivity** F_{res} : $\Delta\rho = -0.060$, 1/4 groups.
- **Gateway centrality** G : held-out $\Delta\text{AUC} = -9 \times 10^{-6}$ on 8,515 held-out episodes.
- **Within-concept closure**: DEV not supported ($b = +0.015$, CI includes 0); held-out reverses sign (−0.079).

The trajectory typology failed the naming rule (DTW-vs-HMM ARI = 0.222); the data supports a continuum, not discrete classes. External recognition (O5) is unrelated to publication outcomes.

4 Strength of Evidence and Caveats

- **Replication:** OPEN confirmed on fresh cohort (Holm $p = 0.048$, marginal) and partially on Frame N (Holm $p = 0.052$). Six non-selection bodies all positive.
- **Coupling:** OPEN_{all} (+0.17) is mechanically coupled; about half is paper count. OPEN_{home} (+0.069) is the decoupled estimate.
- **Practical value:** No forecasting gain over a simple popularity baseline.
- **Reliability:** Split-half $r_{SB} = 0.49$; yearly ICC = 0.37.
- **Heterogeneity:** Life & Environment is weakest. Retained-frontier $I^2 = 0.92$.
- **Causal ambiguity:** The within-concept closure test is null. Openness may reflect intrinsic generality rather than a causal mechanism.

5 Run Statistics

Five iterations; 25 artifacts commissioned, 21 completed, 1 incomplete (Exp11, whose remaining models Exp15 later ran), 3 failed. Review scores: 3, 3, 3, 2, 2 (all blocking). Total ledger spend: \$20.87. The hypothesis moved from gateway fields (iter. 1) through retained frontiers (iter. 2–3) to openness (iter. 4–5), with every intermediate lead closing on larger data. Confidence decreased at each iteration until stabilising at iteration 5.